IBM SkillsBuild AICTE 2026 Internship

Problem Statement No. 41

Power System Fault Detection and Classification

Domain

Electrical Engineering (Machine Learning Project)

The Challenge

Design a machine learning model to detect and classify different types of faults in a power distribution system. Using electrical measurement data (e.g., voltage and current phasors), the model should be able to distinguish between normal operating conditions and various fault conditions (such as line-to-ground, line-to-line, or three-phase faults). The objective is to enable rapid and accurate fault identification, which is crucial for maintaining power grid stability and reliability.

Objective

Build a machine learning model that can:

- Analyze electrical and environmental measurement data (voltage, current, power load, temperature, wind speed) from a power distribution system.
- Classify the type of fault occurring — Line Breakage, Overheating, or Transformer Failure.
- Enable rapid and accurate fault identification to support power grid stability and reliability.
- Provide a deployable, testable model that gives real-time predictions on new fault data.

Dataset

Kaggle: Power System Faults Dataset
https://www.kaggle.com/datasets/ziya07/power-system-faults-dataset

Technology

Use of IBM Cloud Lite services is mandatory.

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